Analytical Chemistry Cumulative Examination
January 21, 2004

Capillary zone electrophoresis is a separation technique that is complementary to electrophoresis. Please answer the following questions about CZE, as it is sometimes called.

1. What is the principle of separation in electrophoresis? CZE is different from "normal" electrophoresis in that electroosmosis contributes to the separation. What is electroosmosis? Does it help or hinder the technique?

2. CZE has also been termed "high performance capillary electrophoresis". Why?

3. How are samples introduced in CZE (give two examples)?

4. What are the difficulties in detection in CZE? How have they been overcome (give at least three examples)? What kind of detection limits would you expect for these detectors?

5. One area where CZE has great promise is in the separation and quantitation of biopolymers. What are the problems in this area? How have they been overcome (give two examples)?

Two useful review articles published in the early days of CZE are: I. M. Warner, *Anal. Chem.*, **60**, 1159A (1988) and M. J. Gordon, et. al., *Science*, **242**, 224 (1988).